Jaime Bruno Cirne de Oliveira

NECTA: Neural Connectivity Time-resolved Analysis

Natal – Rio Grande do Norte, Brazil

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Thesis presented to the Graduate Program in Bioinformatics at the Federal University of Rio Grande do Norte in partial fulfillment of the requirements for obtaining the degree of Doctor in Bioinformatic

Advisor: Prof. Dr^{a}. Kerstin Erika Schmidt

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2026

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Escreva aqui seus agradecimentos.

"It pays to keep an open mind, but not so open your brains fall out."
(Carl Sagan)

ABSTRACT

Escreva seu resumo aqui. Ele deve ser parágrafo único e sem récuo na primeira linha. O resumo deve tratar das informações gerais do trabalho, trazendo tema, objetivo(s), metodologia, principais resultados e considerações finais. Em geral, no resumo não cabem citações e estruturas enumeradas são incomuns. Toda sigla deve ser definida na primeira vez em que é mencionada, inclusive no resumo.

Palavras-chaves: palavra1; palavra2; palavra3; palavra4; palavra5.

ABSTRACT

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Keywords: keyword1; keyword2; keyword3, keyword4; keyword5.

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LIST OF ABBREVIATIONS AND ACRONYMS

HTML	*HyperText Markup Language*
JSON	*JavaScript Object Notation*
REST	*Representational State Transfer*

LIST OF SYMBOLS

Γ	Letra grega Gama
Λ	Lambda
ζ	Letra grega minúscula zeta
$\in$	Pertence

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1 INTRODUCTION

1.1 The Primary Visual Cortex and Mesoscopic Networks

The primary visual cortex has long served as the canonical model for understanding cortical microcircuitry, sensory processing, and network topology. Historically, the architectural and functional parcellation of visual areas—such as Area 17 (A17), Area 18 (A18), and their Transition Zone (TZ) in the feline model—has been extensively mapped using single-unit recordings and static receptive field analyses (Wunderle et al., 2013; Schmidt et al., 2023). These classical approaches have provided the fundamental "structural backbone" of the visual system, detailing how intra- and inter-hemispheric connections form clustered functional communities. However, while static topologies describe the anatomical constraints of neural communication, they obscure the rapid, transient fluctuations in functional integration that are the hallmark of active sensory processing.

1.2 From Static Topologies to the “Dynome”

Cognitive processes and the integration of complex visual stimuli do not rely on fixed pathways, but rather on highly dynamic, time-varying networks. Recognizing this limitation, the neuroscientific community has begun shifting its focus from static connectomics to the exploration of the "Dynome"—the millisecond-scale evolution of functional networks and their temporal dynamics (KOPELL et al., 2014). Capturing the Dynome requires shifting the analytical framework from time-averaged correlations to time-resolved state transitions, often achieved through sliding-window approaches (ALLEN et al., 2014). In this context, observing how network metrics, such as *Small-Worldness* and modularity, evolve over brief temporal windows provides a much more accurate representation of the brain’s real-time functional state.

1.3 Rhythmic Oscillations and Gamma Synchronization

At the mesoscopic scale, rhythmic neural oscillations—specifically in the Gamma frequency band (30–90 Hz)—act as a fundamental mechanism for cortical communication and information routing. Synchronization in the low-gamma band has been established as a core driver for visual feature binding and functional cohesion across the retinogeniculate and early cortical pathways of the cat (NEUENSCHWANDER et al., 2023). Furthermore, the directionality of these high-frequency rhythms dictates hierarchical processing. Theoretical and empirical models propose that distinct frequency bands carry specific feedforward

or feedback information, establishing temporary master-slave relationships, or "Driver-Receiver" dynamics, between distinct cortical areas (VEZOLI et al., 2021). Thus, mapping the directionality of gamma synchronization is crucial for deciphering the causal architecture of the visual cortex.

1.4 The Methodological Bottleneck in Electrophysiology

Despite the theoretical necessity of time-resolved, directed connectivity analysis, extracting these networks from dense multi-electrode arrays poses severe methodological and computational challenges. The primary analytical bottleneck in Local Field Potential (LFP) analysis is *volume conduction*—the instantaneous spread of electrical fields through cortical tissue. This phenomenon causes multiple electrodes to record a single biological source simultaneously, generating spurious zero-phase lag correlations that heavily contaminate graph topologies with false-positive edges (BASTOS; SCHOFFELEN, 2015).

Furthermore, the transition to high-density electrophysiology (e.g., 32-64 channel matrices) creates a "Big Data" computational crisis. Calculating dynamic, non-linear connectivity metrics (such as Partial Directed Coherence or phase-based indices), generating hundreds of surrogate networks for statistical validation, and tracking graph properties over moving time windows inevitably lead to combinatorial explosions. Traditional, single-threaded analytical scripts are inherently incapable of managing such stochastic, multidimensional loads, often resulting in Out-of-Memory (OOM) failures or prohibitive processing times.

Consequently, there is a critical need for a modern, high-performance computing (HPC) framework capable of mitigating volume conduction artifacts through advanced phase-based estimators while leveraging distributed parallel processing to interactively render the mesoscopic Dynome.

2 WORKING HYPOTHESIS AND OBJECTIVES

2.1 Working Hypothesis

Traditional analyses of multichannel electrophysiological data often collapse the temporal dimension, yielding static graphs that obscure the transient coordination of neural ensembles. Furthermore, attempts to extract time-resolved networks are frequently confounded by volume conduction and crippled by the computational intractability of massive stochastic testing.

We hypothesize that integrating highly scalable parallel computing with phase-based, zero-lag immune connectivity estimators (e.g., Imaginary Coherence and Weighted Phase Lag Index) and dynamically optimized causal models (Partial Directed Coherence) will overcome these traditional bottlenecks. By deploying such a framework, it will be possible to decode the "Dynome" of the primary visual cortex—revealing that the anatomical structural backbone (Areas 17, 18, and the Transition Zone) supports highly flexible, time-varying topological reorganizations (such as transient peaks in Small-Worldness) that align with active phases of sensory processing.

2.2 General Objective

The primary objective of this thesis is to develop, validate, and empirically apply **NECTA (Neural Electrophysiological Connectivity Time-resolved Analysis)**, an open-source, high-performance computational framework designed for the interactive exploration and statistical validation of dynamic mesoconnectomes from raw high-density electrophysiological recordings.

2.3 Specific Objectives

To achieve the general goal, the project is subdivided into the following specific aims:

1. **Data Ingestion and Standardization:** To develop a robust pipeline (the *Isaac* ecosystem) capable of converting dispersed legacy electrophysiological binaries (e.g., SPASS format) into standardized, cloud-native, n-dimensional tensors (Zarr/Xarray) using lazy-loading mechanisms.

2. **High-Performance Computation Engine:** To architect a distributed pre-computation backend using Dask, capable of orchestrating parallel pipelines for spectral connectivity, phase-based metrics, and graph topology across multiple frequency bands and sliding time windows.

3. **Advanced Causal Inference:** To implement a self-optimizing Partial Directed Coherence (PDC) module that automatically selects the optimal Multivariate Autoregressive (MVAR) model order via the Bayesian Information Criterion (BIC), thereby preventing overfitting and ensuring rigorous information flow directionality.

4. **Statistical Rigor and Null Modeling:** To embed strict statistical validation protocols, including False Discovery Rate (FDR) corrections applied to phase-randomized surrogate data, and topological null models (Erdös-Rényi and Configuration Models) to validate graph communities.

5. **Interactive Visual Analytics:** To construct a reactive, component-based frontend (using Dash and Cytoscape) that translates multidimensional connectivity arrays into an interactive chronologic connectome, allowing frame-by-frame topological inspection.

6. **Empirical Validation *In Vivo*:** To apply the NECTA framework to an empirical dataset (‘c1608a01‘) obtained from the feline visual cortex (A17/TZ/A18) to characterize the hierarchical directionality of gamma-band synchronization and the temporal evolution of network efficiency during visual stimulation.

3 MATERIALS AND METHODS

3.1 Animal Preparation and Data Acquisition

The empirical validation of the framework utilized the ‘c1608a01‘ dataset (WUNDERLE; ERIKSSON; SCHMIDT, 2013). Multichannel electrophysiological recordings (Local Field Potentials and multi-unit spikes) were obtained from the primary visual cortex of an adult cat. To isolate intrinsic cortical network dynamics from confounding behavioral variables (e.g., saccadic eye movements or fluctuations in conscious attention), the animal was maintained under general anesthesia and paralysis. High-density matrix electrodes (32–64 channels) were implanted spanning Area 17 (A17), the A17/18 Transition Zone (TZ), and Area 18 (A18).

3.2 Data Ingestion Architecture (The Isaac Pipeline)

To process the high-density recordings, the *VisionICeIO* (Isaac) pipeline was developed to act as a bridge between legacy proprietary formats (e.g., SPASS binaries) and modern data structures.

- **Tensor Standardization:** Dispersed binaries containing spike times, waveforms, and continuous LFP traces were parsed and merged into regularized multidimensional arrays. Ragged arrays (e.g., varying spike counts per trial) were standardized using ‘NaN‘ padding.

Zarr and Xarray Encapsulation: *The output is structured as a cloud-native ‘Zarr‘ store. By encapsulating the data within an ‘xarray.Dataset‘, the framework leverages* lazy loading*, allowing the frontend to read metadata trees without exhausting system RAM.

3.3 Distributed Pre-Computation Engine (HPC)

Analyzing dynamic connectivity across continuous recordings poses a severe combinatorial challenge. NECTA addresses this through a distributed pre-computation engine orchestrated by **Dask**.

- **Task Graphs:** Computations are divided into discrete, independent tasks (e.g., computing coherence for a specific frequency band and time window) and mapped across a cluster of worker nodes.

- **Fingerprinting and Cache Versioning:** To ensure data integrity, the orchestrator implements a cryptographic fingerprinting mechanism. Any modification to analytical parameters (e.g., altering variance thresholds or VAR criteria) alters the global hash, safely invalidating obsolete ‘Zarr‘ caches.

3.4 Spectral Connectivity and Phase Analysis

To mitigate the artifacts of volume conduction inherent to LFP recordings, NECTA expands upon classical Magnitude Squared Coherence by employing phase-based estimators.

- **Analytical Signal Extraction:** Raw LFP signals are processed using the Hilbert Transform to yield the analytical signal, granting instantaneous access to both phase $\phi(t)$ and amplitude $A(t)$.
- **Imaginary Coherence (iCoh) & Weighted Phase Lag Index (wPLI):** The framework computes ‘iCoh‘ to strictly isolate interactions with non-zero phase delays. Furthermore, ‘wPLI‘ is calculated by weighting the phase differences by the magnitude of the imaginary component of the cross-spectrum, providing a highly robust metric against uncorrelated noise and instantaneous spatial spread.

3.5 Directed Functional Connectivity (PDC)

To infer causal interactions and hierarchical information flow, NECTA implements an advanced Partial Directed Coherence (PDC) pipeline.

1. **Stationarity Filtering:** LFP windows exceeding a predefined variance threshold are identified as artifacts and discarded, preserving the assumptions of autoregressive modeling.
2. **Automated Order Selection:** Instead of imposing a static model order, the Dask workers dynamically fit Multivariate Autoregressive (MVAR) models using Ordinary Least Squares (OLS). The algorithm iterates through orders (up to `max_order=20`) and selects the optimal p that minimizes the Bayesian Information Criterion (BIC), balancing model fit against parametric complexity.
3. **Frequency Transformation:** Autoregressive coefficients A_r are transformed into the frequency domain $A(f)$ using Fast Fourier Transforms accelerated via ‘Numba‘ (FastMath), yielding the normalized PDC matrices.

3.6 Graph Theory and Statistical Validation

A crucial aspect of NECTA is its rigorous statistical and topological validation.

- **Surrogate Data & FDR Correction:** For connectivity edges (e.g., PDC), significance is established by generating 100 phase-randomized surrogate datasets. Edge-wise p-values are computed and subjected to False Discovery Rate (FDR) correction (Benjamini-Hochberg procedure, $\alpha = 0.05$). Non-significant connections are mathematically zeroed.
- **Topological Metrics:** The framework extracts core graph properties including Modularity (Louvain heuristic), Efficiency, and Assortativity (the tendency of nodes to connect with similar-degree nodes).
- **Small-Worldness (σ) and Null Models:** NECTA calculates the small-world coefficient by comparing the empirical graph's clustering and path length against theoretical Erdös-Rényi models. Furthermore, macro-topology is validated against Configuration Models (via double-edge swapping) to ensure that the detected community structures preserve empirical degree distributions.

3.7 Interactive Visualization Dashboard

The final tier of NECTA is a Single Page Application (SPA) built with 'Dash' and 'Cytoscape'. The interface does not process raw signals; it performs on-the-fly slicing of the pre-computed 'Zarr' cache using 'xarray.sel()'. Changes in the global state (e.g., sliding the temporal window) trigger instantaneous callbacks that serialize exact data slices into JSON, seamlessly re-rendering complex Connectome Matrices, Temporal Integration scatter plots, and anatomical Information Flow graphs.

4 RESULTS

4.1 Baseline Characterization and the Static Backbone

To establish a baseline structural framework, static functional connectivity was initially evaluated across the entire recording session. Applying Magnitude Coherence in the Low Gamma band (30–59 Hz) with a proportional density threshold of 15

4.2 Hierarchical Information Flow (Directed Connectivity)

To decode the causal relationships within this structural backbone, Partial Directed Coherence (PDC) was applied using the BIC-optimized MVAR models. The static PDC analysis in the Low Gamma band revealed a highly asymmetrical and hierarchical information flow across the visual cortex. By calculating the Net Flow (Out-Degree minus In-Degree) for each region, A17 emerged as the dominant persistent *Driver* (Source) of the network, presenting a strong positive net flow (+1.10). In contrast, A18 functioned as the primary *Receiver* (Sink), exhibiting a robust negative net flow (-1.36). These results establish a clear and dominant directional pathway extending from A17 through the TZ and terminating in A18 (A17 → TZ → A18), consistent with canonical feedforward processing hierarchies.

4.3 Dynamic Topology and Temporal Integration

The core capability of NECTA lies in capturing the "Dynome"—the transition from static networks to chronological connectomes. To investigate this, connectivity was recalculated using a sliding-window approach ($W = 500$ ms, 50

The dynamic analysis revealed that the network's topological efficiency is not static. Instead, it responds organically to visual processing phases. The network achieved its absolute peak in *Small-Worldness* specifically during the moving grating stimulus phase. This transient optimization was driven primarily by a rapid surge in local clustering, indicating that the anesthetized cortical network actively reorganizes into a highly efficient state to process dynamic visual inputs.

4.4 The "Dynome" Evolution and Modulatory Nodes

Extracting frame-by-frame snapshots (Connectome Filmstrip) of the dynamic Low Gamma Coherence network illustrated a stable rhythmic organization over time. The hubs located in A17 and the TZ maintained cohesive clustering, providing a robust temporal context. Crucially, evaluating the *Net Flow Evolution* over time uncovered a complex temporal interplay that static analysis obscures. While A17 consistently acted as a "Volatile Source" and A18 as a "Stable Sink", the Transition Zone (TZ) exhibited unique behavior. The TZ functioned as a *Dynamic Modulator*, actively switching its directional role (from sink to source, or vice versa) precisely at the onset of visual stimulation. This highlights the framework's capacity to dissect the mesoconnectome chronologically, revealing dynamic hierarchical relays that coordinate sensory integration.

5 DISCUSSION

5.1 From Static Maps to the Visual Dynome

Historically, the functional organization of the primary visual system has been delineated through the static mapping of receptive fields and anatomical inter-hemispheric tracing (Wunderle et al., 2013; Schmidt et al., 2023). While these classical methods successfully outline the spatial constraints and functional columns of the cortex, they inherently capture only a time-averaged "structural backbone." Cognitive processing and the integration of moving stimuli, however, depend on millisecond-scale state transitions within neural ensembles—a temporal dimension defined as the *Dynome* (KOPELL et al., 2014).

The application of the NECTA framework to the ‘c1608a01‘ dataset provides empirical proof of this dynamic nature. By utilizing a continuous sliding-window approach, we demonstrated that even under conditions of general anesthesia and paralysis—where confounding conscious top-down modulations are suppressed—the visual cortex undergoes rapid, stimulus-locked topological reorganizations. The observation that the network reaches transient peaks in *Small-Worldness* precisely during the moving grating stimulus demonstrates that macroscopic network efficiency is an active, responsive property rather than a fixed anatomical trait.

5.2 Mitigating Electrophysiological Confounders

A profound challenge in high-density LFP connectomics is the pervasive artifact of volume conduction, wherein a single electrical source is instantaneously registered by multiple neighboring electrodes (BASTOS; SCHOFFELEN, 2015). If unaccounted for, this phenomenon generates spurious zero-phase lag correlations, heavily biasing the resulting graph topology with false-positive edges.

NECTA addresses this methodological bottleneck architecturally. By employing the Hilbert Transform to extract analytical signals, the framework calculates phase-based indices such as the Weighted Phase Lag Index ($wPLI$) and Imaginary Coherence ($iCoh$), which are mathematically immune to instantaneous spatial spread. This algorithmic refinement ensures that the dynamic topologies extracted—such as the discrete communities mapping to A17, TZ, and A18—reflect true underlying physiological synchronization rather than electrical crosstalk.

5.3 Gamma Synchronization and Causal Hierarchies

Previous work from our group has established the cardinal role of gamma-band synchronization in functional routing and cohesion within the retinogeniculate and cortical pathways of the cat (NEUENSCHWANDER et al., 2023). NECTA extends this foundation by mapping the causal directionality of these high-frequency rhythms.

By applying Partial Directed Coherence (PDC) optimized dynamically via the Bayesian Information Criterion (BIC), we eliminated the bias of arbitrary model-order selection. The resulting hierarchy in the Low Gamma band (A17 acting as the persistent Driver, A18 as the Receiver) strongly corroborates current theories of predictive coding and feedforward information routing (VEZOLI et al., 2021). Furthermore, the identification of the Transition Zone (TZ) as a "Dynamic Modulator" that reverses its causal flow upon stimulus onset highlights the necessity of time-resolved directional tools. Static PDC would have merely averaged this reversal, missing a critical operational mechanism of the visual cortex.

5.4 Overcoming the Computational Scale

Translating the theoretical concepts of the Dynome into practical analytical outputs requires overcoming severe computational barriers. Generating stochastic multivariate null models (e.g., surrogate data phase-randomization and configuration models) across continuous sliding windows triggers a combinatorial explosion that typically crashes single-threaded analytical scripts. By orchestrating these routines through a Dask-distributed High-Performance Computing (HPC) backend and utilizing ‘xarray‘ and ‘Zarr‘ for lazy-loading operations, NECTA effectively eliminates the "Big Data" electrophysiology bottleneck. This pipeline not only democratizes complex graph-theoretical analysis but allows researchers to interactively slice and visualize multidimensional connectomes in real-time, bridging the gap between raw signals and system-level dynamic insights.

6 CONCLUSÃO

Escreva suas conclusões, limitações do seu trabalho, contribuições, trabalhos futuros, etc.

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